| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **4** | **(i)** | | $\frac{dy}{dx} = 1 - \frac{1}{(x-2)^2}$ | **M1** | **1.1a** | Attempt to differentiate with one term correct | |
| | | | | **A1** | **1.1** | Correct derivative | |
| | | | $1 - \frac{1}{(x-2)^2} = 0$ at stationary points | **M1** | **1.1a** | | |
| | | | $x - 2 = \pm 1$ so $x = 1, \quad 3$ | **A1** | **2.2a** | Both values of $x$ | |
| | | | $(1, -5)$ $(3, -1)$ | **A1** | **1.1** | Both values of $y$ - ft *their* $x$ | |
| | | | | **[5]** | | | |

| Question | | | Answer | Marks | AOs | Guidance | |
|---|---|---|---|---|---|---|---|
| 4 | (ii) | | $\frac{d^2y}{dx^2} = \frac{2}{(x-2)^3}$ | M1 | 1.1a | | |
| | | | $x = 3 \quad \frac{d^2y}{dx^2} > 0$ (2) so minimum | A1 | 2.4 | | |
| | | | $x = 1 \quad \frac{d^2y}{dx^2} < 0$ (-2) so maximum | A1<br>[3] | 2.4 | OR Allow consideration of gradient either side of stationary point for **M1**<br>Correct gradients above and below each tp **A1**<br>Correct convincing conclusions (possibly with sketches ) **A1**<br>$x$ f($x$) $x$ f'(x)<br>0.5 0.56 2.5 -3.00<br>0.6 0.49 2.6 -1.78<br>0.7 0.41 2.7 -1.04<br>0.8 0.31 2.8 -0.56<br>0.9 0.17 2.9 -0.23<br>1 0.00 3 0.00<br>1.1 -0.23 3.1 0.17<br>1.2 -0.56 3.2 0.31<br>1.3 -1.04 3.3 0.41<br>1.4 -1.78 3.4 0.49<br>1.5 -3.00 3.5 0.56 | |
| 4 | (iii) | | $x = 2$ | B1<br>[1] | 1.2 | | |

| Question | | | Answer | Marks | AOs | Guidance | |
|---|---|---|---|---|---|---|---|
| **4** | **(iv)** | | $x > 2$ | **A1**<br><br>**[1]** | **2.2a** | | FT *their* (iii) if region is to right of *their* $x$ value |